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# PAM

## MIDI to CV Converter

Mono · 1V/oct · 12-bit DAC · Gate + S-Trig

### FEATURES

- 12-bit MCP4725 DAC — accurate 1V/oct pitch CV
- Gate output — compatible with most analogue synths
- S-Trig output — onboard jumper selects VTRIG or STRIG
- MIDI channel selection — omni or specific channel
- Root note setting — match any keyboard octave convention
- Transpose — shift pitch +/-12 semitones
- On-screen calibration — trim with DVM via CV jack lead
- Portamento/glide — smooth pitch slides between notes
- Note range filter — 7 presets for keyboard splits
- MIDI activity indicator — blue dot flashes on MIDI receive
- EEPROM storage — all settings survive power cycles
- Colour TFT display — live note, gate, CV readout




**Made in Belfast, Northern Ireland**

User Guide v3.1 · Alan G Watkins 2026

## WHAT IS PAM?

PAM converts MIDI note messages into analogue control voltages (CV) and gate signals, allowing you to play analogue synthesisers from any MIDI keyboard, sequencer or DAW. It follows the 1 volt per octave standard used by most analogue and modular synths.

## CONNECTIONS

| JACK / CONNECTOR | FUNCTION | NOTES |
|---|---|---|
| **MIDI IN (DIN5)** | MIDI input | Standard 5-pin DIN MIDI |
| **CV>** | Pitch CV output | 0-5V, 1V/oct |
| **GATE>** | Gate/Trig output | Selected by onboard jumper |
| **DC 7-12V** | Power input | Centre positive, regulated |

## GATE TYPE JUMPER (VTRIG / STRIG)

PAM has a 3-pin jumper on the board labelled GATE TYPE to select the gate signal type. The centre pin (pin 2) is common. Fit the jumper to select:

| JUMPER POSITION | SIGNAL TYPE | USE WITH |
|---|---|---|
| **Pins 1-2 (right) DEFAULT** | VTRIG — active high 5V gate | Most modern synths |
| **Pins 2-3 (left)** | STRIG — active low trigger | Moog, early Korg |

PAM is shipped with the jumper in the VTRIG (pins 1-2) position.

## MIDI CHANNEL

PAM can respond to all MIDI channels (OMNI) or a single specific channel.
Press button A to cycle: OMNI -> CH1 -> CH2 -> ... -> CH16 -> OMNI
The current channel is shown in the top right of the display.

## ROOT NOTE SETTING

Different keyboards use different MIDI note numbers for their lowest C.
Press button C to cycle the root note convention to match your keyboard:

| | |
|---|---|
| **C1 (MIDI 24)** | Most modern keyboards and DAWs — default setting |
| **C2 (MIDI 36)** | Some keyboards label their lowest C as C1 but send MIDI 36 |
| **C3 (MIDI 48)** | Older keyboards, some Yamaha models |
| **C4 (MIDI 60)** | Some bass synths and older instruments |

Setting is saved to EEPROM and restored on power up.

## TRANSPOSE

Press button B to shift pitch up one semitone per press (-12 to +12). Saved to EEPROM.

## MAIN DISPLAY

| | |
|---|---|
| NOTE | Current MIDI note name and number. Shows --- when no note held. |
| dot | Blue dot top-right of header flashes on MIDI activity. |
| GATE | GREEN = gate high (note held) GREY = gate low (no note) |
| CV | Current DAC output voltage and raw count. Yellow bar shows level. |
| TRAN | Current transpose. Shows none at zero. |
| ROOT | Keyboard root: C1 / C2 / C3 / C4. Orange when not default. |
| GLIDE | Current portamento setting. MAGENTA when active, grey when off. |
| STRIG | MAGENTA = S-Trig active GREY = S-Trig idle |

## BUTTONS — NORMAL MODE

| | |
|---|---|
| A | Cycle MIDI channel (OMNI -> 1 -> 2 -> ... -> 16 -> OMNI) |
| A hold | PANIC — all notes off, DAC returns to C1 voltage |
| B | Transpose — shift pitch up one semitone per press (-12 to +12) |
| B hold | Open help screen |
| C | Cycle root note convention (C1 / C2 / C3 / C4) |
| C hold | Open settings (portamento + calibration) |

## BUTTONS — SETTINGS SCREEN

| | |
|---|---|
| A | Move cursor between menu items |
| B | Change value of selected item |
| C | Enter / confirm (calibrate only) |
| C hold | Exit settings |

## SETTINGS — NOTE RANGE PRESETS

| | |
|---|---|
| **FULL** | All notes 0-127 — no filtering (default) |
| **LOW OCT** | C1 to B2 — lower two octaves |
| **MID LOW** | C2 to B3 — mid-bass range |
| **MID** | C3 to B4 — middle keyboard range |
| **MID HIGH** | C4 to B5 — upper mid range |
| **HIGH** | C5 and above — upper keyboard |
| **BASS** | C1 to B1 — single bass octave only |

## BUTTONS — CALIBRATION MODE

| | |
|---|---|
| **A** | Coarse up +10 counts |
| **A hold** | Coarse down -10 counts |
| **B** | Fine up +1 count |
| **B hold** | Fine down -1 count |
| **C** | Advance to next step / Save and exit on final step |
| **C hold** | Exit calibration without saving |

## CALIBRATION PROCEDURE

Device is factory calibrated but there may be occasions you want to compensate for tolerance in your 5V power supply. Plug a jack lead into the CV> socket and touch your DVM probes to the tip and sleeve of the free end.

### STEP 1 Root voltage — C1 = 1.000V

- Hold C button to enter calibration mode.
- The DAC outputs C1. Gate goes high to sound the synth.
- Use A / A-hold for coarse adjustment (+/-10 counts).
- Use B / B-hold for fine adjustment (+/-1 count).
- Adjust until DVM reads exactly 1.000V.
- Press C to advance to step 2.

### STEP 2 Scale factor — C2 = 2.000V

- The DAC now outputs C2 (one octave up from C1).
- Touch DVM probes to jack tip and sleeve again.
- Adjust using the same A and B buttons.
- Adjust until DVM reads exactly 2.000V.
- Press C to save calibration and exit.

## QUICK SETUP GUIDE

1. Set jumper to VTRIG (pins 1-2) or STRIG (pins 2-3) as needed.
2. Connect MIDI IN to your keyboard or sequencer.
3. Connect CV> to your synth pitch CV input.
4. Connect GATE> to your synth gate/trigger input.
5. Power on PAM. CV output stabilises at 1V (C1) immediately.
6. Press C to cycle root note until notes play at correct octave.
7. Press B to transpose if fine pitch offset is needed.
8. Hold C to enter calibration and trim voltages with DVM via CV jack.

## SPECIFICATIONS

| | |
|---|---|
| **Processor** | ESP32-WROOM-32E |
| **DAC** | MCP4725 / ZC576300 12-bit I2C |
| **CV output range** | 0 - 5V (1V/oct) |
| **CV resolution** | 12-bit (4096 steps) |
| **Gate output** | VTRIG 5V active high or STRIG active low (jumper select) |
| **MIDI input** | Standard DIN5, 31250 baud, optoisolated |
| **Display** | ST7735S 1.8" colour TFT 160x128 |
| **Power input** | 7 - 12V DC, centre positive |
| **Note priority** | Last note, 8 note stack |
| **Root note options** | C1 (MIDI 24) / C2 (MIDI 36) / C3 (MIDI 48) / C4 (MIDI 60) |
| **Transpose range** | -12 to +12 semitones |
| **MIDI channels** | Omni or single channel 1-16 |
| **Settings storage** | EEPROM — survives power cycles |
| **Firmware** | v3.1 |

Specifications subject to change. Visit www.mindburner.co.uk for latest information.